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RESEARCH ARTICLE

Beyond Positive Similarity Metrics: Leveraging Negative Co-Occurrence in Recommender Systems

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ABSTRACT Given the growing amount of online information and the lack of effective tools to extract relevant content for individuals, recommender systems (RSs) have emerged as highly efficient means to provide personalized recommendations to users. Collaborative filtering (CF) is a popular recommendation technique that suggests items to the target user based on the ratings provided by similar users in the system. However, the performance of CF, specifically memory-based CF, relies on choosing the convenient similarity metric. An optimal choice of the similarity metric will lead to better recommendations. Despite extensive studies on RSs in the past decade, research on similarity metrics in RSs has only recently gained attention. This study investigates 37 similarity metrics that incorporate negative co-occurrence for implicit ratings. We conduct a comparative analysis of these metrics against various real-world datasets using well-known performance measures. Our study provides insights into effective similarity metrics and their performance in relation to data characteristics. Specifically, our results reveal that while the Russel-Rao similarity metric consistently outperforms other similarity metrics across various datasets in terms of precision, there is no superior similarity metric across all datasets and performance metrics.

INDEX TERMS Recommender systems, collaborative filtering, implicit ratings, similarity metrics, negative co-occurrence.

I. INTRODUCTION

The growing amount of online information highlights the importance of recommender systems (RSs) [1], [2]. These tools provide users with useful item suggestions, thus assisting them in various decision-making processes [3], [4]. Collaborative filtering (CF), a recommendation technique, typically relies on measuring the similarity between users or items [5], [6], [7]. Over the years, researchers have used traditional RSs based on CF in various contexts, such as Peer-to-Peer systems [8], [9], [10] and Blockchain [11]. However, it is noteworthy that most of the works that deal with CF do not delve into similarity metrics, besides a few exceptions [12], [13], [14]. Interestingly, the field of biology employs a far wider range of similarity and distance metrics to analyze the relationships between biological objects [15]. This diversity is not commonly reflected in RSs, nor in machine learning more broadly. This disparity motivated our study, as we believe importing and adapting these metrics could bring significant value to machine learning in general and CF in the field of RS specifically. Based on this insight, our study conducts extensive simulations to assess different similarity metrics utilizing negative co-occurrence for implicit ratings, which are rarely used in traditional CF contexts. This assessment aims to assist researchers, practitioners, users, and developers in selecting the most suitable similarity metric for their RSs. We have performed extensive simulations on seven real-world datasets, namely MovieLens-100K, MovieLens-1M, MovieLens-10M, Jester, Epinions, Electronics, and MovieTweetings, employing various performance metrics.

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Hence, our study makes the following notable contributions:

- Surveying the literature to identify the most popular similarity metrics used in RSs.
- Presenting similarity metrics that use implicit ratings to provide a valuable reference for researchers.
- Testing the performance of a RS using different similarity metrics against different datasets.
- Providing a reference for RS developers to select the appropriate similarity metrics from a broad collection of possible metrics.
- Demonstrating Smart Adaptive Recommendations (SAR)'s versatility across various contexts and improving its effectiveness by incorporating advanced similarity metrics beyond traditional ones like Lift and Jaccard.

A detailed methodology was used to select the most relevant articles for the review. The search engines used in the study included IEEE Xplore Digital Library, ACM Digital Library, Science Direct, Semantic Scholar, PapersWithCode, ResearchGate, and Google Scholar. Various combinations of keywords, such as RSs, similarity metrics, collaborative filtering, distance metrics, and binary similarity were used in the search.

This research was limited to the period between 2004 and 2024 to ensure that the analyzed work was relevant and covered the entire topic.

The structure of this article is as follows: Section II provides an overview of the most important concepts and terms related to CF for recommendations. Section III discusses previous related work on comparative analysis of similarity metrics. Section IV highlights the similarity metrics for implicit ratings in RSs that have been used in recent years. The performance evaluation of the similarity metrics is presented in Section V, where we detail the tools and datasets used for the simulations and provide a set of evaluation indicators. This section concludes with a comparison and discussion of the results. Finally, Section VI concludes the paper and provides an outlook on future work and perspectives.

II. RECOMMENDER SYSTEMS

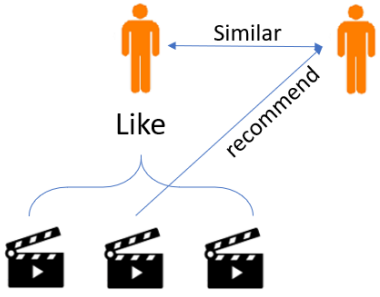
Recommendation engines use several techniques to suggest products or items to users [16]. These techniques include CF, content-based (CB) filtering, knowledge-based techniques, and hybrid methods. As the name indicates, CB filtering is a technique based on analyzing the content and descriptions of items and using this information to predict their relevance to the user. It is particularly useful when there are not enough item ratings. On the other hand, knowledge-based recommender systems generate recommendations by using detailed information about users and products to determine which options best meet user needs. They often involve interactive processes to refine recommendations [16], [17]. Finally, hybrid methods combine two or more filtering techniques and/or various RSs to achieve better results. In this section, we focus on the basic principles behind CF.

CF is the process of filtering items using the opinions of other users. CF is based on correlations in users' rating histories [18] and aims to predict unknown ratings. This technique examines the user-item rating matrix to determine whether users or items are similar. On the basis of this analysis, in essence, the system recommends items that are popular with users who are similar to the target user. CF techniques can be categorized into two categories: memory-based CF and model-based CF [5], [7], [19].

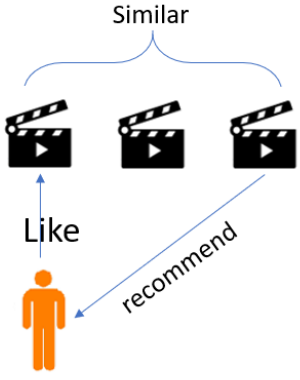
Memory-based CF relies on user or item neighborhoods defined by similarity [16].

This technique includes two approaches: user-based CF and item-based CF. We assume that there is a specific target user for whom the recommendations are intended. With user-based CF, we identify the users who are most similar to the target user, also known as neighbors. We then recommend the items that the neighbors liked the most to the target user. For item-based CF, we search for items that are very similar to those liked and highly rated by the target user, and then recommend these items to them.

Figure 1 illustrates the principles of user-based CF and item-based CF. As shown in Figure 1a, the user-based approach suggests items to the target user that similar users liked. Likewise, for item-based CF, as depicted in Figure 1b, items similar to those liked by the target user are identified and suggested.



(a) user-based CF



(b) item-based CF

FIGURE 1. User-based and item-based CF.

When dealing with large-scale systems that involve numerous users and items, memory-based CF becomes impractical for real-world applications due to its limitations in handling large volumes of data. Model-based CF is employed as an alternative to address this issue. The model-based approach aims to build a machine-learning model that captures the behavior of users and uses this information to make predictions about their preferences. Before making recommendations, model-based CF requires a pre-training phase to determine the best model parameters, as described in [20]. Once the learning phase is completed, the model will be able to predict user ratings quickly and effectively.

III. RELATED WORK

A similarity metric is a mathematical tool used in RSs to determine the degree of resemblance or association between two or more users or items. The metric calculates the similarity between users or items and makes it possible to find similar people or items that a customer might be interested in.

In the following, we present related work from previous studies on similarity metrics.

The study in [21] provides an overview and classification of various distance and similarity metrics used to compare probability density functions linked to nominal-type histograms. The authors use hierarchical clustering to classify distances and similarity metrics and provide a correlation coefficient to identify related measures.

In [22], the authors propose a matrix factorization framework that incorporates social regularization. The authors use vector space similarity (VSS) and Pearson correlation coefficient (PCC) to calculate the similarity between users. The results showed that PCC-based methods perform slightly better than VSS-based methods.

In [23], the authors study the effect of the choice of distance measurement on the efficiency of the K-nearest neighbor (KNN) classifier. They evaluated the performance of KNN using numerous distance measures on real datasets, both in the presence and absence of noise. The results of the experiments show that the performance of the KNN classifier is strongly influenced by the choice of the distance measure.

In their study [24], the authors examined the effectiveness of various similarity measures in a CF setting, with a focus on data sparsity. They compared the performance of traditional similarity metrics such as PCC, cosine correlation, Jaccard similarity, constrained Pearson correlation coefficient (CPCC), sigmoid Pearson correlation coefficient (SPCC), Euclidean distance, and the City-Block similarity metric. The authors conclude that the choice of similarity metric is an important factor in the accuracy of recommendations and that the Minkowski metric is a good choice in scenarios with sparse data.

Further, the work in [25] focused on three structural similarity measures similar to the Jaccard index and provided a theoretical analysis and an experimental comparison of these metrics in the context of CF recommendation.

The authors in [7] present the theoretical foundations of 13 well-known similarity metrics used in CF, followed by an experimental comparison on three different datasets. Furthermore, the experimental evaluation was carried out for user-based CF and item-based CF. The results show that improved triangle rating (ITR) and improved PCC weighted with ratings preferences behavior (IPWR) demonstrate the best performance for user-based CF, while adjusted mutual information (AMI) is the best for item-based CF, suggesting the absence of a single similarity metric suitable for both approaches.

To the best of our knowledge, existing works lack a comprehensive survey of various similarity metrics - beyond traditional ones- in the context of memory-based CF. Although some works, such as [24], analyze similarity metrics in the context of providing recommendations, they focus on a limited set of traditional metrics. Thus, our study introduces a broad selection of similarity metrics into the field of recommender systems, with a focus on negative co-occurrence, which is often overlooked due to its less intuitive nature. Specifically, we focus on memory-based CF using implicit ratings, where interactions are represented in unary form. Our evaluation involves testing these metrics on the SAR model, which has primarily been examined with traditional similarity metrics. Unlike other studies, we select a diverse range of real-life datasets with varying characteristics to evaluate the similarity metrics across different contexts, allowing us to derive valuable conclusions for researchers in the field.

IV. SIMILARITY METRICS WITH NEGATIVE CO-OCCURRENCE

Explicit feedback refers to the ratings provided by users of RSs, usually ranging from 0 to 5 stars, stating their preferences and opinions regarding the recommended items or content. Usually, a high rating means that the user likes the item, while a low rating means that the user dislikes it.

Another form of feedback is implicit feedback, which is very common and easier to collect. Implicit feedback is unary in type, e.g., click, buy, watch, listen, etc. For example, a click signals a positive preference, and the absence of a click does not necessarily mean a dislike [16]. In this section, we provide an overview of the most common similarity metrics using implicit ratings.

In this work, we focus on item-based CF with implicit ratings. Therefore, inspired by [25] and [26], we express similarity metrics in terms of operational taxonomic units (OTUs). For two items i and j encoded as binary feature vectors with n features, we can derive Table 1. The presence of a feature in the binary vector is denoted as 1, and its absence as 0.

The meaning of the notations used is given in Table 2.

Since the implicit ratings matrix is binary, we use the following co-occurrence matrix to compute the values of parameters a, b, c, and d, where co_{ij} is the number of positive

TABLE 1. OTUs expression of binary instances i and j.

j/i	1 (Presence)	0 (Absence)	$\sum$
1 (Presence)	$a\ (=j \bullet i)$	$b\ (=j \bullet \bar{i})$	$a+b$
0 (Absence)	$c\ (=\bar{j} \bullet i)$	$d\ (=\bar{j} \bullet \bar{i})$	$c+d$
$\sum$	$a+c$	$b+d$	$n=a+b+c+d$

co-occurrences between items i and j.

	Items			
	i_1	i_2	$\cdots$	i_n
Items				
i_1	$co_{i_1i_1}$	$co_{i_1i_2}$	$\cdots$	$co_{i_1i_n}$
i_2	$co_{i_2i_1}$	$co_{i_2i_2}$	$\cdots$	$co_{i_2i_n}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
i_n	$co_{i_ni_1}$	$co_{i_ni_2}$	$\cdots$	$co_{i_ni_n}$

In this case, for two items i and j, as in [23], a, b, c, and d are expressed as follows in Eq. (1):

$$\begin{aligned} a &= co_{ij} \\ b &= co_{jj} - a \\ c &= co_{ii} - a \\ d &= n - (a+b+c) \end{aligned} \tag{1}$$

Similarity metrics with negative co-occurrence use the parameter "d", which indicates how often an item i does not occur together with another item j, as shown in Table 2 [27], [28], [29], [30], [31], [32], [33]. Traditional metrics like cosine similarity or PCC rely on positive co-occurrence "a", as it is more easily interpretable and suggests a natural relationship between items. However, negative co-occurrence captures shared absences, which is less intuitive and thus less commonly studied. For implicit ratings, interactions are unary. Therefore, understanding the absence can be as informative as understanding the positive interaction.

In Table 3, we present 37 binary similarity and distance metrics that incorporate negative co-occurrence, identified through a review of widely cited and commonly used metrics across various domains [26], [27]. Each metric is accompanied by its corresponding mathematical formula, type (similarity metric (S) or distance metric (D)), and respective range.

V. PERFORMANCE EVALUATION

In this section, we present the experimental setting in which we evaluate the similarity metrics by examining their impact on the quality of recommendations. The code for the simulations is available and can be accessed through the repository.[1]

[1]https://github.com/KHHD99/SM-IN-RS

TABLE 2. Meaning of the notations used.

Symbol	Meaning
a	Represents the number of features where both i and j have a value of 1 indicating a positive co-occurrence.
b	Represents the number of attributes where $i = 0$ and $j = 1$ indicating an absence mismatch for i.
c	Represents the number of attributes where $i = 1$ and $j = 0$ indicating an absence mismatch for j.
d	Represents the number of attributes where both i and j have a value 0 indicating a negative co-occurrence.
$a+d$	Represents the total number of co-occurrences between i and j.
$b+c$	Represents the total number of mismatches between i and j
$n=a+b+c+d$	Represents the total sum of the quantities in the 2×2 table.

A. RECOMMENDER SYSTEMS IMPLEMENTATION TOOLS

This section presents the tools we have tested to select the best for our simulations.

1) *TuriCreate*[2] is a Python machine-learning library made by Apple that simplifies the process of creating add-on recommendations, object recognition, image classification, image similarity, or activity classification systems.
2) *LensKit* is a set of Python tools for experimenting and studying RSs [34]. It is an open-source toolkit that provides support for training, executing, and evaluating recommendation algorithms.
3) *Crab* is a Python recommendation engine that typically combines conventional information filtering techniques with Python scientific programs (NumPy, SciPy, and Matplotlib) [35].
4) *Microsoft Recommenders* is available as an open source repository on GitHub [36], [37]. At the time of writing, the repository is the most popular for RS on GitHub, with 18.6k stars. Microsoft Recommenders supports both memory-based and model-based models. All methods and tools were developed in Python, and the implementations are in the form of Jupyter notebooks.[3]

We analyze and compare these recommendation tools. In the following, we present our findings:

- TuriCreate is a good tool, but it is difficult to first compile it from source code and then include the similarity metrics we want to study.
- The core of TuriCreate is written in C++, and the tests are written in Python.
- TuriCreate requires a variety of models and packages to run correctly.
- TuriCreate cannot be compiled directly on Windows.

[2]https://apple.github.io/turicreate/docs/
[3]https://jupyter.org/

TABLE 3. Similarity and distance metrics with negative co-occurrence.

Metric	Formula	S/D	Range
1_Sokal-Michener	$\frac{a+d}{a+b+c+d}$	S	$[0, 1]$
2_Sokal-Sneath-II	$\frac{2(a+d)}{2(a+d)+b+c}$	S	$[0, 1]$
3_Sokal-Sneath-IV	$\frac{\frac{a}{(a+b)}+\frac{a}{(a+c)}+\frac{d}{(b+d)}+\frac{d}{(c+d)}}{4}$	S	$[0, 1]$
4_Sokal-Sneath-V	$\frac{ad}{(a+b)(a+c)(b+d)(c+d)^{0.5}}$	S	$[0, 1]$
5_Pearson-I	$\alpha^2 = \frac{n(ad-bc)^2}{(a+b)(a+c)(b+d)(c+d)}$	S	$[-1, 1]$
6_Pearson-II	$(\frac{\alpha^2}{n+\alpha^2})^{\frac{1}{2}}$	S	$[-1, 1]$
7_Pearson-III	$(\frac{\rho}{n+\rho})^2$ where $\rho = \frac{ad-bc}{\sqrt{(a+b)(a+c)(b+d)(c+d)}}$	S	$[-1, 1]$
8_Pearson-Heron-I	$\frac{ad-bc}{\sqrt{(a+b)(a+c)(b+d)(c+d)}}$	S	$[-1, 1]$
9_Pearson-Heron-II	$\cos(\frac{\pi\sqrt{bc}}{\sqrt{ad}+\sqrt{bc}})$	S	$[-1, 1]$
10_Baroni-Urbani-Buser-I	$\frac{\sqrt{ad}+a}{\sqrt{ad}+a+b+c}$	S	$[-1, 1]$
11_Baroni-Urbani-Buser-II	$\frac{\sqrt{ad}+a-(b+c)}{\sqrt{ad}+a+b+c}$	S	$[0, 1]$
12_Forbes-I	$\frac{na}{(a+b)(a+c)}$	S	$[0, \frac{n}{a}]$
13_Forbes-II	$\frac{na-(a+b)(a+c)}{n.min(a+b,a+c)-(a+b)(a+c)}$	S	$[-1, 1]$
14_Yule-Q-S	$\frac{ad-bc}{ad+bc}$	S	$[-1, 1]$
15_Yule-w	$\frac{\sqrt{ad}-\sqrt{bc}}{\sqrt{ad}+\sqrt{bc}}$	S	$[-1, 1]$
16_Tarantula	$\frac{\frac{a}{a+b}}{\frac{c}{c+d}} = \frac{a(c+d)}{c(a+b)}$	S	$[0, 1]$
17_Ample	$\frac{\lvert\frac{a}{a+b}\rvert}{\lvert\frac{c}{c+d}\rvert} = \lvert\frac{a(c+d)}{c(a+b)}\rvert$	S	$[0, 1]$
18_Rogers-Tanimoto	$\frac{a+d}{a+d+2(b+c)}$	S	$[0, 1]$
19_Faith	$\frac{a+0.5d}{a+b+c+d}$	S	$[0, 1]$
20_Gower-Legendre	$\frac{a+d}{a+0.5(b+c)+d}$	S	$[0, 1]$
21_Inner Product	$a + d$	S	$[0, 1]$
22_Russell-Rao	$\frac{a}{a+b+c+d}$	S	$[0, 1]$
23_Tarwid	$\frac{na-(a+b)(a+c)}{na+(a+b)(a+c)}$	S	$[-1, 1]$
24_Dennis	$\frac{ad-bc}{\sqrt{n(a+b)(a+c)}}$	S	$[-\frac{\sqrt{n}}{2}, \frac{\sqrt{n}}{2}]$
25_Gower	$\frac{a+d}{\sqrt{(a+b)(a+c)(b+d)(c+d)}}$	S	$[-1, 1]$
26_Stiles	$\log_{10}(\frac{n(\lvert ad-bc\rvert-\frac{n}{2})^2}{(a+b)(a+c)(b+d)(c+d)})$	S	$[-\infty, +\infty]$
27_Fossum	$\frac{n(a-0.5)^2}{(a+b)(a+c)}$	S	$[0, \frac{(n-0.5)^2}{n}]$
28_Disperson	$\frac{ad-bc}{(a+b+c+d)^2}$	S	$[-1, 1]$
29_Hamann	$\frac{(a+d)-(b+c)}{a+b+c+d}$	S	$[-1, 1]$
30_Michael	$\frac{4(ad-bc)}{(a+d)^2+(b+c)^2}$	S	$[-1, 1]$
31_Peirce	$\frac{ab+bc}{ab+2bc+cd}$	S	$[0, 1]$
32_Eyraud	$\frac{n^2(na-(a+b)(a+c))}{(a+b)(a+c)(b+d)(c+d)}$	S	$[-1, 1]$
33_Yule-Q-D	$\frac{2bc}{ad+bc}$	D	$[0, 2]$
34_Mean-Manhattan	$\frac{b+c}{a+b+c+d}$	D	$[0, 1]$
35_Vari	$\frac{b+c}{4(a+b+c+d)}$	D	$[0, +\infty]$
36_Shape-Difference	$\frac{n(b+c)-(b-c)^2}{(a+b+c+d)^2}$	D	$[0, +\infty]$
37_Pattern-Difference	$\frac{4bc}{(a+b+c+d)^2}$	D	$[0, +\infty]$

- LensKit for Python needs a special environment.
- Crab is an old project; the last modification is from 2012. It is also written in Python 2.7.
- Microsoft Recommenders is a new tool and has been recently updated (The most recent update was a few months ago at the time of writing).
- Microsoft Recommenders is written in Python and can work with the latest machine-learning libraries.
- Microsoft Recommenders can also be installed on UNIX, OS, and Windows.

After studying these recommendation tools, we decided to investigate similarity metrics with Microsoft Recommenders.

B. SMART ADAPTIVE RECOMMENDATIONS ALGORITHM

SAR is an algorithm provided by Microsoft [36], [37]. It is a fast, scalable, and adaptable algorithm that generates personalized recommendations based on the user's transaction history and item descriptions.

From the end user's perspective, SAR offers certain advantages. It is easier to explain to non-technical RS users. It also performs well in the cold-start item recommendation settings by using item data to develop a regression model.

We choose this CF algorithm as a basis to test the performance of similarity metrics. The two similarity metrics, Lift and Jaccard, have already been implemented in the SAR algorithm.

C. DATASETS

To examine the impact of the chosen similarity metrics on RSs, we evaluate them using the SAR algorithm on seven well-known datasets. Namely, MovieLens-100k, MovieLens-1M, MovieLens-10M, Jester, MovieTweetings, Electronics, and Epinions.

- GroupLens MovieLens[4] (100k, 1M and 10M): These datasets of movie ratings of different sizes were collected by GroupLens Research in different time periods [38].
- Jester[5]: This dataset contains 4.1 million continuous ratings (from -10.00 to +10.00) for 100 jokes from 73,496 people. The data was collected between April 1999 and May 2003 [39].
- Epinions[6]: This is a dataset that contains ratings and trust values collected from Epinions.com, a general consumer review site, founded in 1999 [40].
- MovieTweetings[7]: This is a dataset that contains movie reviews from well-structured tweets on X formerly known as Twitter. This dataset is the result of the study carried out in [41].
- Electronics[8]: This is a dataset from the Amazon Reviews Repository that is used to suggest items to consumers based on their previous reviews of other products in the e-Commerce category.

[4]https://grouplens.org/datasets/movielens/
[5]https://grouplens.org/datasets/jester/
[6]https://www.kaggle.com/datasets/masoud3/epinions-trust-network
[7]https://www.kaggle.com/datasets/tunguz/movietweetings
[8]https://www.kaggle.com/datasets/pritech/ratings-electronics?select=ratings_Electronics.csv

Table 4 below summarizes some basic statistics on these datasets.

TABLE 4. Basic statistics of the datasets: ratings, users, items, rating range, trust links, and sparsity.

Datasets	Ratings	Users	Items	Range	Trust	Sparsity
MovieLens 100K	100000	943	1682	[1, 5]	No	0%
MovieLens 1M	1000209	6040	3900	[1, 5]	No	0%
MovieLens 10M	10000054	69878	10677	[1, 5]	No	0%
Jester	1829974	24983	100	[−10, 10]	No	27.747%
Electronics	7824482	4201696	476002	[1, 5]	No	0%
Movie-Tweetings	906831	70783	37337	[1, 10]	No	0.031%
Epinions	598329	49289	139738	[1, 5]	Yes	0%

D. SIMULATION ENVIRONMENT

All experiments were conducted on a high-performance computing system running Rocky Linux, equipped with an Intel(R) Xeon(R) Platinum 8276 CPU @ 2.20GHz with 1.5TB RAM and an NVIDIA A100-SXM4-80GB GPU.

Regarding the experimental protocol, we conduct a 4-fold cross-validation for each dataset. Since we are focusing on implicit feedback and working with well-known datasets that originally contained explicit ratings, we convert the explicit ratings to implicit ratings for consistency. In our experiments, we evaluate various *top_k* values: $top_k = 1$, $top_k = 3$, $top_k = 5$, $top_k = 7$, and $top_k = 10$. We also extend our investigation beyond these values to explore cutoffs up to $top_k = 100$.

Based on our simulations, we found no significant difference in the performance between $top_k = 1, 3, 7$ and $top_k = 10$ as shown in Figures 2. The mean and the median values of MAE, P@k, and R^2 across datasets exhibit stable and consistent performance across different top k values, with only a slight increase for some extreme outliers, as clearly illustrated in Figure 3.

Therefore, we present the results for $top_k = 10$ for the MovieLens, Epinions, Jester, MovieTweetings, and Electronics datasets. This choice balances performance and computational efficiency as it provides a meaningful number of recommendations without the computational overhead associated with high *top_k* values. Additionally, $top_k = 10$ aligns with common practices in recommender systems research.

E. PERFORMANCE INDICATORS

To evaluate the performance, we initially chose the following set of widely used performance metrics. These include error metrics that measure prediction accuracy and ranking metrics that assess the relevance of the recommendation list [42], [43], [44]:

- **MAE:** MAE is a measure of the difference in errors computed as follows:

$$MAE = \frac{\sum_{(u,i)\in T} |r_{ui} - \hat{r}_{ui}|}{|T|} \tag{2}$$

where T represents the test set, $|T|$ represents the test set's cardinal, r_{ui} represents the actual rating given by user u to item i, and $\hat{r}_{ui}$ represents the predicted rating of user u to item i.

- **Root mean squared error (RMSE):** It represents the standard deviation of the prediction errors.

$$RMSE = \sqrt{\frac{\sum_{(u,i)\in T} \left(r_{ui} - \hat{r}_{ui}\right)^2}{|T|}} \tag{3}$$

- **Mean average precision at k (MAP@k):** MAP@k evaluates the algorithm's general accuracy rate by the average precision.

$$MAP@k = \frac{\sum_{i=1}^{k} precision(i) \times rel(i)}{k} \tag{4}$$

where rel(i) equals 1 if j_i is relevant; otherwise, rel(i) equals 0. j_i is an item in $< j_1, j_2, \ldots, j_k >$ which is the sorted list of recommended items of length k for a given user.

- **Normalized discounted cumulative gain at k (NDCG@k):** NDCG@k is a normalization of the DCG score by the IDCG value. The NDCG@k's value will always be between 0 and 1.

$$NDCG@k = \frac{DCG@k}{IDCG@k} \tag{5}$$

where

$$IDCG@k = \sum_{i=1}^{k} \frac{rel(i)}{\log_2(i+1)} \tag{6}$$

$$DCG@k = \sum_{i=1}^{k} \frac{1}{\log_2(i+1)} \tag{7}$$

and k is the length of the recommendation list.

- **P@k:** The precision at k quantifies the positioning of relevant items in the recommended list.

$$Precision@k = \frac{|Relevant_Items \cap Topk_Items|}{k} \tag{8}$$

where *Relevant_Items* denotes the list of items with which users actually interacted and *Topk_Items* refers to the recommended list.

- **Recall at k (R@k):** The Recall at k refers to the proportion of relevant items discovered in the recommendation list.

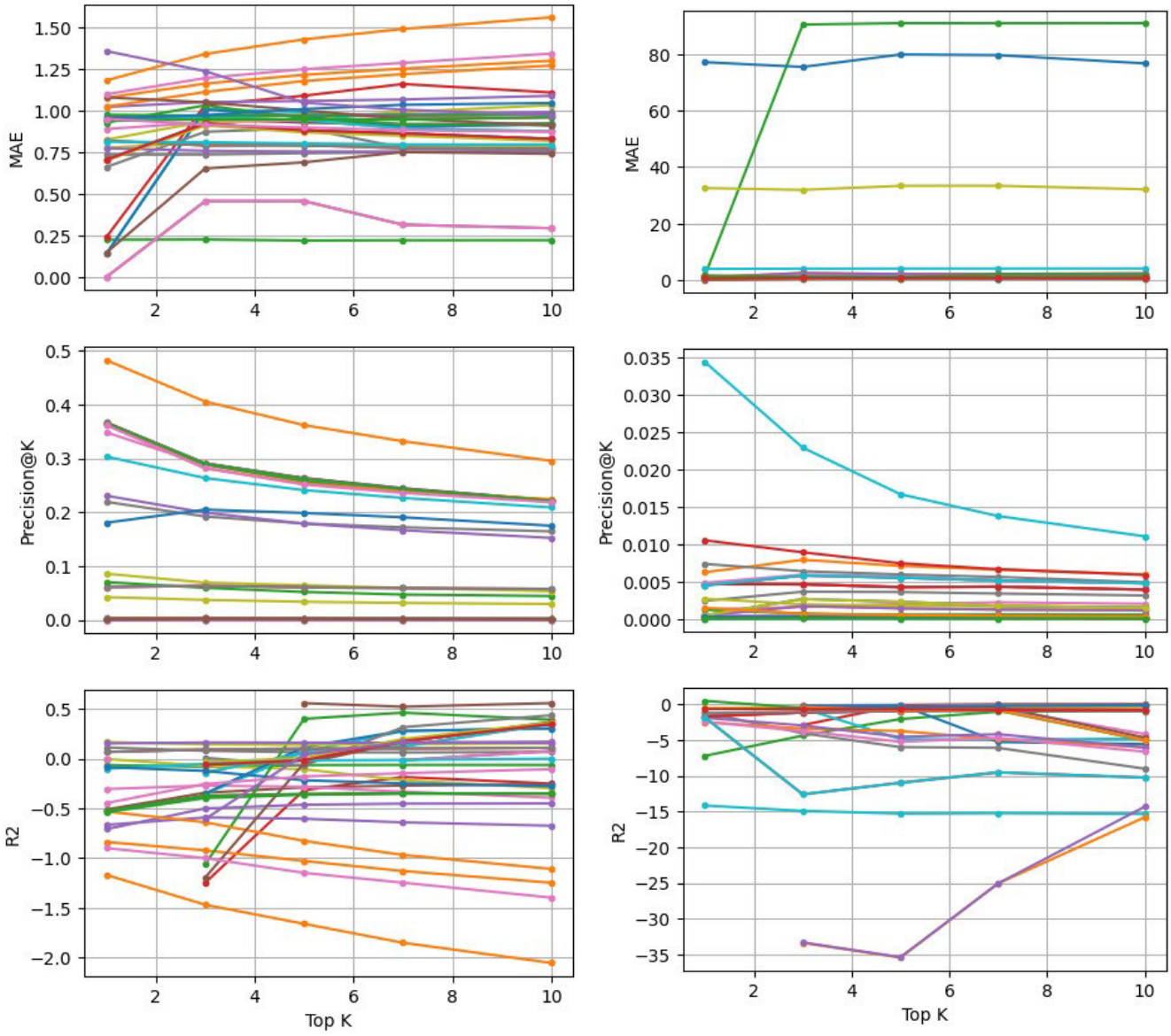
$$Recall@k = \frac{|Relevant_Items \cap Topk_Items|}{|Relevant_Items|} \tag{9}$$

- **F1-score at k (F1@k):** F1 score is the harmonic mean of precision and recall. Thus, it is computed as follows:

$$F1@k = \frac{2 \times (P@k \times R@k)}{P@k + R@k} \tag{10}$$

- **R^2:** It is an indicator used in statistics to judge the quality of linear regression.

$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum_{(u,i)\in T} \left(r_{ui} - \hat{r}_{ui}\right)^2}{\sum_{(u,i)\in T} (r_{ui} - \bar{r})^2} \tag{11}$$



(a) MovieLens-100K dataset

(b) Electronics dataset

FIGURE 2. Evolution of Mean absolute error (MAE), Precision at k (P@k), and R-squared (R^2) for all similarity metrics across different *top_k* values.

where *RSS* and *TSS* are the residual sum of squares and the total sum of squares, respectively, and $\bar{r}$ is the mean of the ratings.

- **Explained Variance:** Also called "explained variation", it is used to quantify the discrepancy between actual and predicted data.

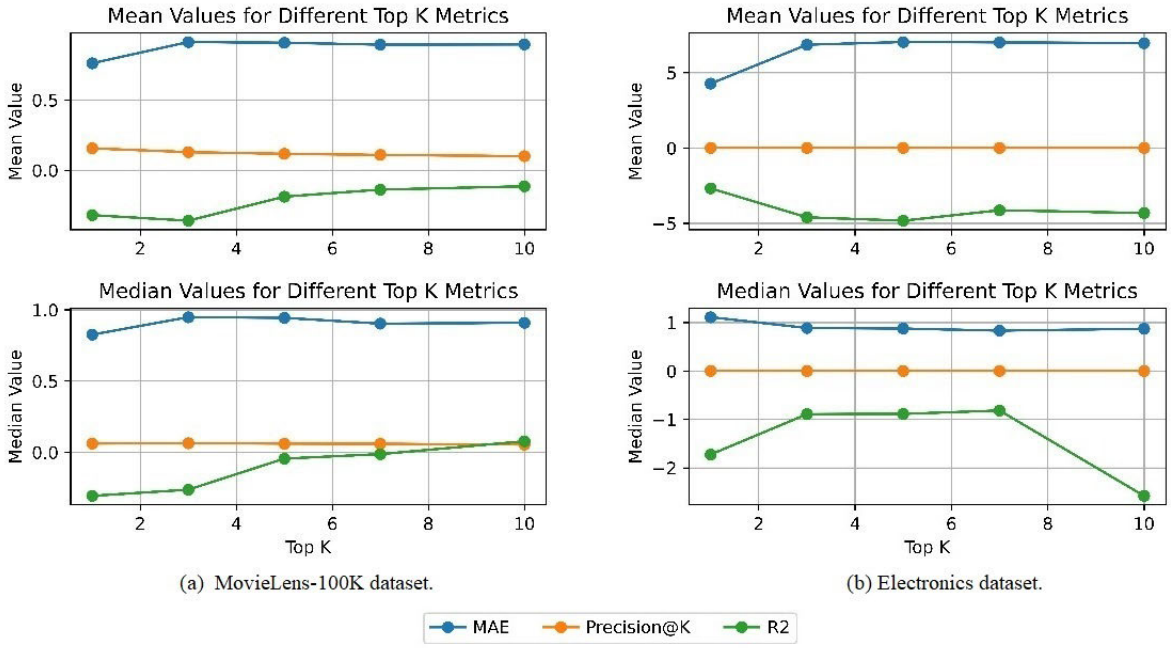
$$ExpVar = 1 - \frac{\text{Var}(r_{ui} - \hat{r}_{ui})}{\text{Var}(r_{ui})} \tag{12}$$

where *Var* is the variance and r_{ui} and $\hat{r}_{ui}$ are the original and predicted ratings of item i by user u, respectively.

Figure 4 shows the correlation between the evaluation metrics for the MovieLens and Epinions datasets. We observed a high correlation between certain evaluation indicators. Specifically, we find positive correlations between RMSE and MAE, between R^2 and the explained variance, and among NDCG@k, MAP@k, P@k, R@k, and F1@k. Consequently, we choose only one indicator from each group to evaluate the effect of similarity metrics on RSs. The chosen indicators are MAE, R^2, and P@k.

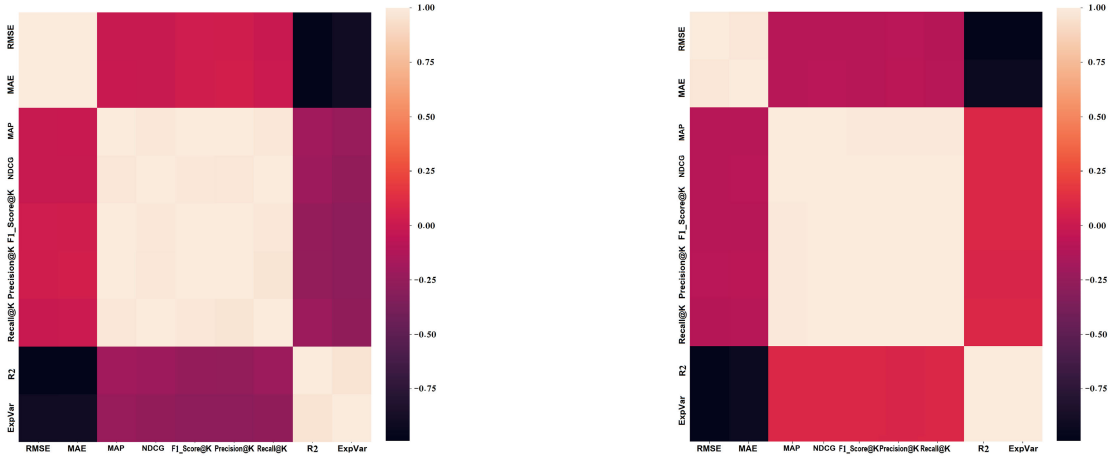
F. EVALUATION RESULTS

In this subsection, we present the results of the comparison between similarity metrics for all datasets. The lower value of MAE and higher value of P@k indicate better performance. For R^2, the closer the value is to 1, the better the performance.



(a) MovieLens-100K dataset. (b) Electronics dataset.

FIGURE 3. Mean and median values of MAE, P@k, and R^2 for different *top_k* values.



(a) MovieLens 100K dataset (b) Epinions dataset

FIGURE 4. Correlation between evaluation metrics.

The following tables and figures show the top-performing similarity metrics. The best results for each performance indicator are in bold. The complete performance tables can be found in the Appendix.

TABLE 5. Top MAE, P@k, and R^2 scores for MovieLens-100K.

Similarity metric	MAE	P@k	R^2
Russell-Rao	1.271247	**0.295413**	-1.108979
Sokal-Sneath-IV	**0.221852**	0.044379	-0.063575
Shape-Difference	0.740371	0.001352	**0.558813**

Based on MovieLens-100k results in Table 5 and Figure 5, we can make the following observations. Similarity metrics Sokal-Sneath-IV, Tarantula, and Ample have the lowest MAE, indicating good prediction performance. However, they also demonstrate a low P@k score and a mediocre R^2, as seen in Table 13 in the Appendix. On the other hand, Russell-Rao exhibits the highest P@k score, indicating its ability to retrieve relevant information in the top-10 items. Meanwhile, the similarity metric Shape-difference has the highest R^2.

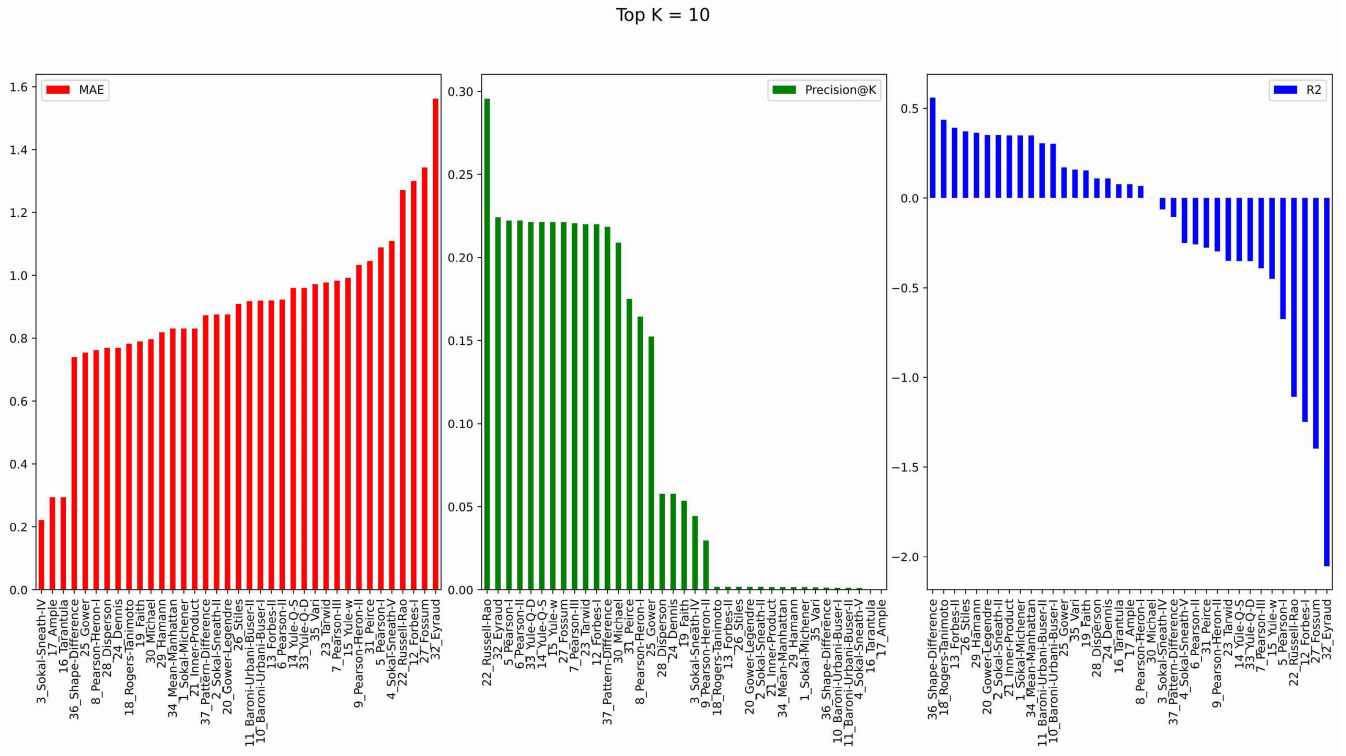


FIGURE 5. MAE, P@k, and R^2 for MovieLens-100K.

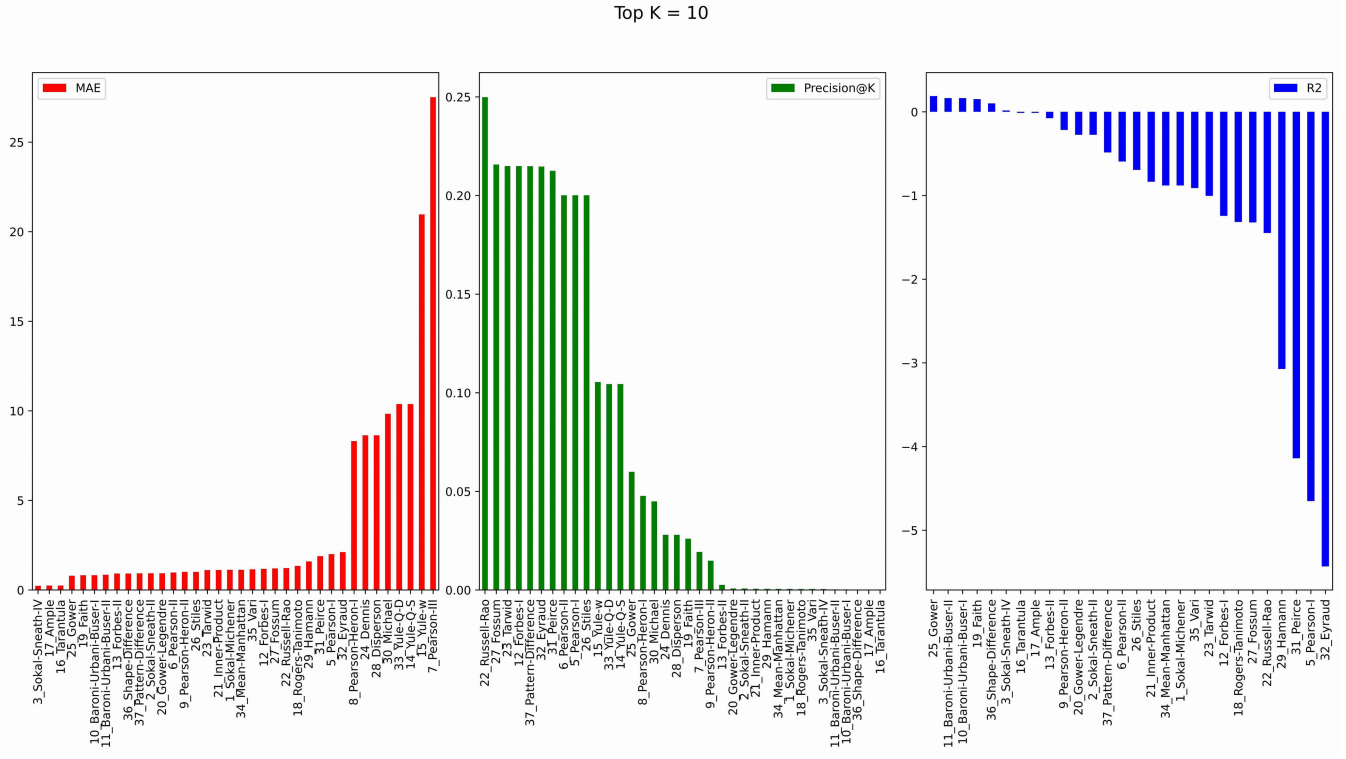


FIGURE 6. MAE, P@k, and R^2 for MovieLens-1M.

Upon examination of the MovieLens-1M results in Table 6 and Figure 6, we notice that the similarity metrics Sokal-Sneath-IV, Tarantula, and Ample have the lowest MAE and poor R^2 and P@k. On the other hand, Dennis, Disperson,

TABLE 6. Top MAE, P@k, and R^2 scores for MovieLens-1M.

Similarity	MAE	P@k	R^2
Russell-Rao	1.218886	**0.249822**	-1.448582
Sokal-Sneath-IV	**0.229868**	0.000438	0.015780
Gower	0.793196	0.059884	**0.187176**

Pearson-Heron-I, Yule-Q-D, Yule-Q-S, Michael, Pearson-III, and Yule-w have unfavorable R^2 scores, as seen in Table 14 in the Appendix. We did not show the R^2 scores for these metrics in Figure 6 because the corresponding values are too large and would hide details of other similarity metrics since they have smaller values. These similarity metrics also have mediocre P@k while Gower exhibits the best performance in terms of R^2. Russell-Rao, Fossum, Tarwid, Forbes-I, and Pattern-Difference have the best P@k values. In particular, Russell-Rao performs well for both MovieLens-100k and MovieLens-1M.

TABLE 7. Top MAE, P@k, and R^2 scores for MovieLens-10M.

Similarity metric	MAE	P@k	R^2
Russell-Rao	1.243323	**0.248301**	-1.430222
Tarantula	**0.111757**	2.73373E-05	0.021107
Ample	**0.111757**	2.73373E-05	0.021107
Pearson-I	0.781337	0.000136	**0.391587**

For the MovieLens-10M dataset, by analyzing the results in Table 7 and Figure 7, the similarity metrics Tarantula and Ample outperform the other metrics in terms of MAE. However, their P@k and R^2 scores are unfavorable. Russell-Rao, Eyraud, Fossum, and Pattern-Difference exhibit the highest P@k, indicating effective retrieving of relevant items in the top-10 results. Other similarity measures, on the other hand, seem to be ineffective for movie recommendations. For example, Yule-Q-D, Yule-Q-S, Pearson-Heron-I, Dennis, Dispersion, Yule-w, Michael, and Pearson-III all achieve a large negative R^2, as seen in Table 15 in the Appendix.

TABLE 8. Top MAE, P@k, and R^2 scores for Jester.

Similarity metric	MAE	P@k	R^2
Tarwid	**2.092937**	0.013037	0.251526
Dennis	3.168473	**0.076414**	-0.022993
Disperson	3.168621	**0.076414**	-0.022992
Pearson-II	2.104398	0.013030	**0.252725**

For the Jester dataset, according to the results in Table 8 and Figure 8, Tarwid, Pearson-II, Stiles, and Sokal-Sneath-V achieve the lowest MAE scores and highest R^2 values, but also obtain unfavorable P@k scores. Dennis and Disperson score the highest in terms of P@k. Moreover, Peirce, Yule-Q-S, and Yule-Q-D score very unfavorably in terms of R^2, as noted in Table 16 in the Appendix, indicating that the use of these similarity metrics results in models failing to explain the Jester data.

TABLE 9. Top MAE, P@k, and R^2 scores for Epinions.

Similarity metric	MAE	P@k	R^2
Pearson-I	1.937170	**0.009066**	-1.790181
Tarantula	**2.2245E-16**	1.89681E-05	**0.000000**
Ample	**2.2245E-16**	1.89681E-05	**0.000000**

For the Epinions dataset, we observe from both Table 9 and Figure 9 that Pearson-I achieves the highest scores in terms of P@k, along with Peirce, Michael, and Russell-Rao. Although their MAE and R^2 scores are less favorable, these results indicate they still provide good relevant item suggestions. Tarantula and Ample both present abnormally low MAE and P@k and a null R^2, indicating a probable overfitting problem. On the other hand, Pearson-III, Baroni-Urbani-Buser-II, Pearson-Heron-II, Gower, and Pearson-Urbani-Buser-I demonstrate good MAE scores. At the same time, Yule-Q-S, Yule-Q-D, Hamann, Mean-Manhattan, Sokal-Michener, Inner Product, Eyraud, Faith, Dennis, Disperson, and Michael have notably large negative R^2 scores -as shown in Table 17 in the Appendix- that will hide details about other similarity metrics with low values, so we do not include them in Figure 9.

TABLE 10. Top MAE, P@k, and R^2 scores for MovieTweetings.

Similarity metric	MAE	P@k	R^2
Baroni-Urbani-Buser-I	**0.189984**	0.000247	**0.133843**
Russell-Rao	6.860635	**0.027965**	-16.950614

For MovieTweetings, we notice in Table 10 that, among the metrics evaluated, Baroni-Urbani-Buser-I has the lowest MAE score and the highest R^2 value, though its value remains low. It also shows an extremely low P@k, the highest P@k value belonging to Russell-Rao. On the other hand, as seen only in Table 18 in the Appendix, other similarity metrics such as Innerproduct, Mean-Manhattan, Sokal-Michener, Yule-Q-S, Yule-Q-D, Disperson, Dennis, Tarwid, Eyraud, and Michael showcase poor results in terms of R^2.

TABLE 11. Top MAE, P@k, and R^2 scores for Electronics.

Similarity metric	MAE	P@k	R^2
Sokal-Sneath-IV	**0.447297**	0.000645	-0.938746
Sokal-Sneath-V	0.670059	0.000159	**-0.004092**
Russell-Rao	4.105240	**0.011112**	-15.345139

For Electronics, we observe, from Table 11, that Sokal-Sneath-IV has the lowest MAE, and that Sokal-Sneath-V outperforms the other metrics in terms of R^2. However, the value is negative, thus signaling a model that does not capture the dataset's characteristics. On the other hand, Russell-Rao is the top-performing metric once again for P@k, though the value is still relatively low. Moreover, as observed in Tables 18 and 19, Russell-Rao and Dennis achieve poor

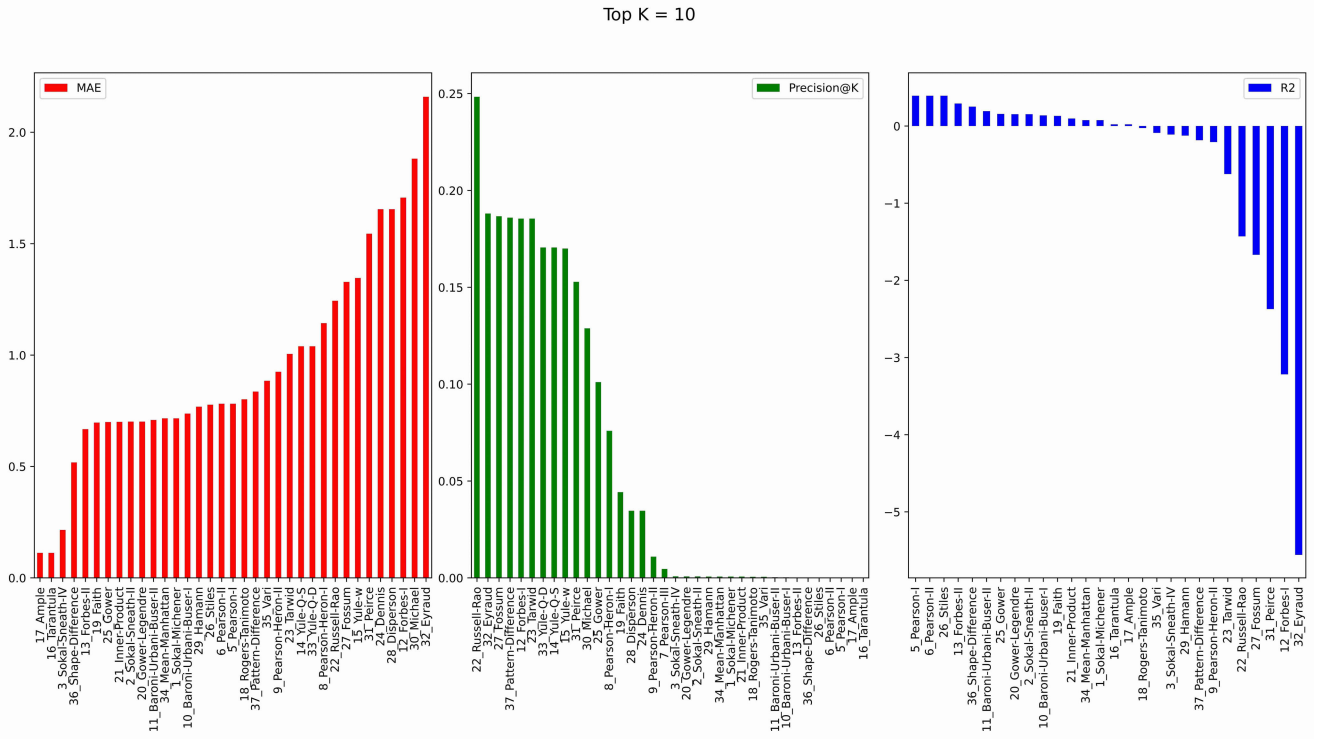


FIGURE 7. MAE, P@k, and R^2 for MovieLens-10M.

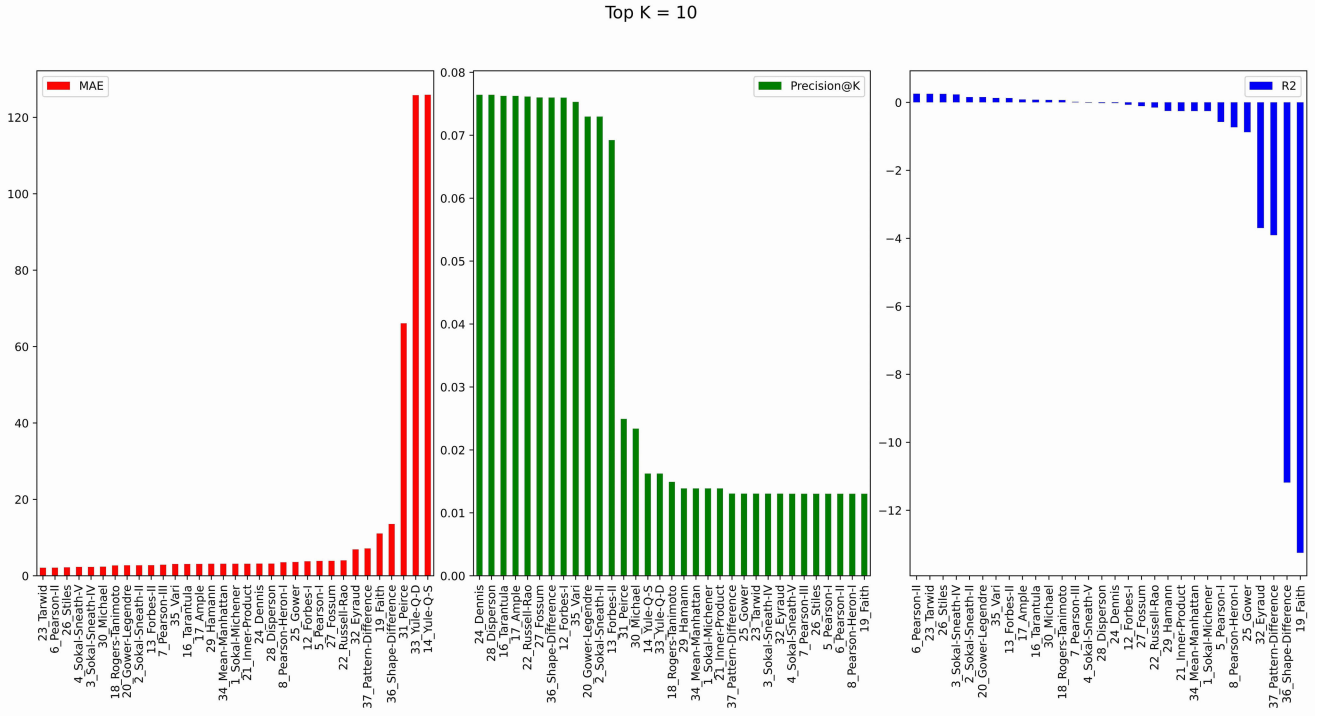


FIGURE 8. MAE, P@k, and R^2 for Jester.

results in terms of R^2. As mentioned above, we do not include Eyraud, Tarwid, and Forbes-II in the Figures because their large values would obscure details about the smaller values of the other similarity metrics.

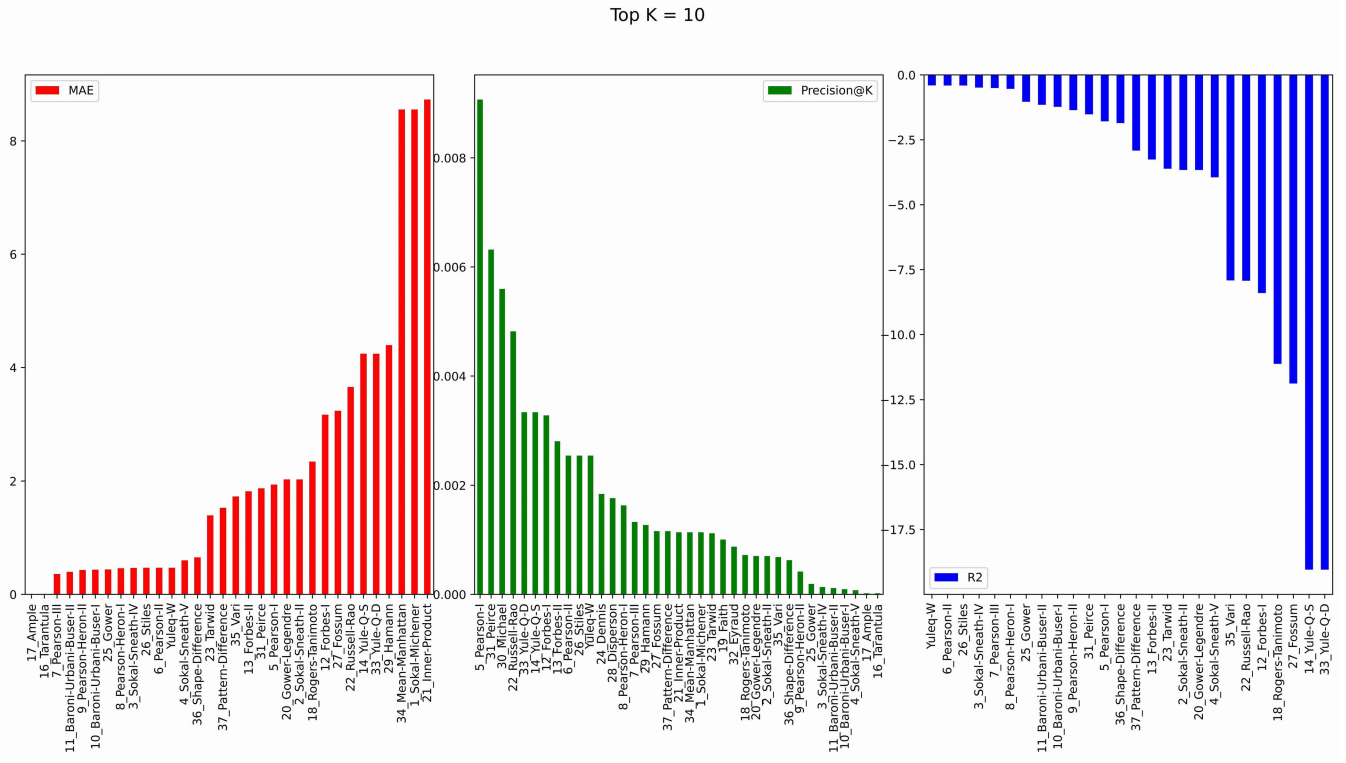


FIGURE 9. MAE, P@k, and R^2 for Epinions.

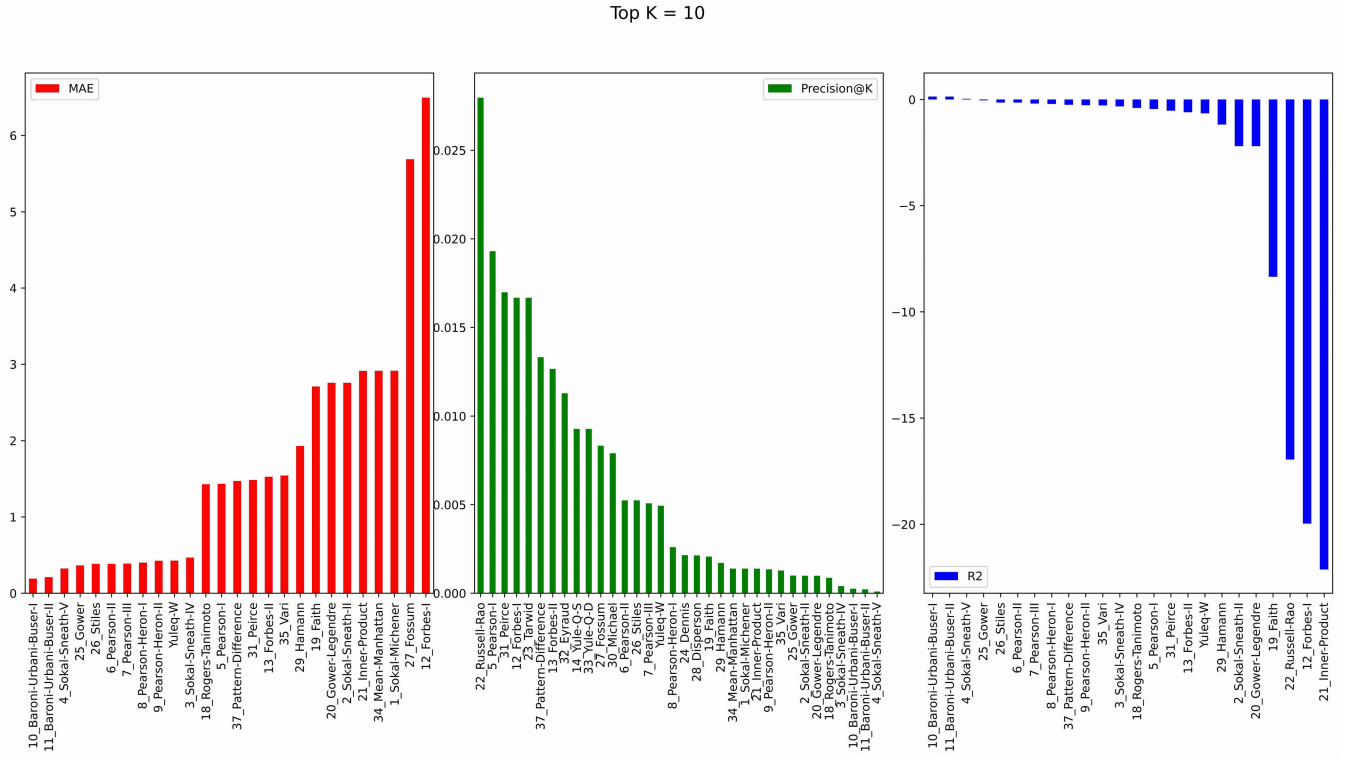


FIGURE 10. MAE, P@k, and R^2 for MovieTweetings.

G. DISCUSSION

The analysis across the MovieLens-100k, 1M, and 10M datasets reveals consistent patterns in the performance of similarity metrics. In all three datasets, the Sokal-Sneath-IV, Tarantula, and Ample metrics consistently demonstrated the lowest MAE, highlighting their strong predictive accuracy.

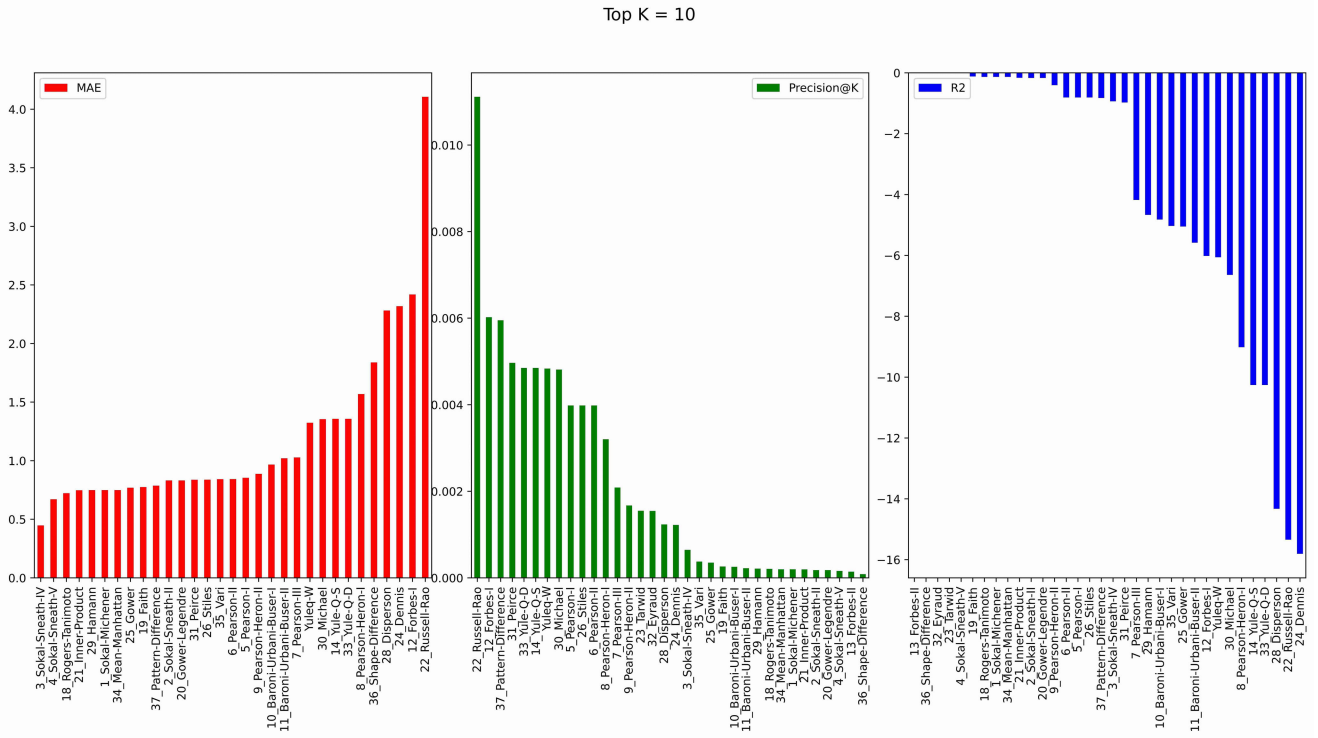


FIGURE 11. MAE, P@k, and R^2 for Electronics.

TABLE 12. Classification of the similarity metrics according to the performance evaluation.

Dataset	MAE (strong)	MAE (poor)	P@k (strong)	P@k (poor)	R^2 (strong)	R^2 (poor)
MovieLens-100k	Sokal-Sneath-IV	Eyraud, Fossum	Russell-Rao	Ample, Tarantula	Shape-Difference	Eyraud
MovieLens-1M	Sokal-Sneath-V	Pearson-III, Yule-w	Russell-Rao	Ample, Tarantula	Gower	Pearson-III, Yule-w
MovieLens-10M	Ample, Tarantula	Pearson-III	Russell-Rao	Ample, Tarantula	Pearson-I	Pearson-III
Jester	Tarwid	Yule-Q-D, Yule-Q-S	Dennis, Disperson	Faith	Pearson-II, Tarwid	Peirce, Yule-Q-D, Yule-Q-S
Epinions	Tarantula, Ample	Dennis, Disperson, Michael	Pearson-I	Tarantula, Ample	Yule-w, Pearson-II	Michael
Movie-Tweetings	Baroni-Urbani-Buser-I	Michael	Russell-Rao	Sokal-Sneath-V	Baroni-Urbani-Buser-I, Baroni-Urbani-Buser-II	Michael
Electronics	Sokal-Sneath-IV	Forbes-II	Russell-Rao	Shape-Difference	Sokal-Sneath-V	Dennis

However, this performance was not mirrored in their P@k and R^2 scores, which were generally mediocre, indicating that these metrics may prioritize prediction over relevance and model fit. A notable trend is the exceptional performance of the Russell-Rao metric in P@k across all MovieLens datasets. This metric consistently achieved the highest P@k scores, demonstrating its effectiveness in identifying the most relevant items within the top-10 recommendations. This suggests that Russell-Rao is particularly adept at relevance-based tasks, despite not necessarily excelling in predictive accuracy or overall model fit and data explanation. This underscores the importance of selecting similarity metrics based on the specific goal of the recommendation system, whether it is prioritizing predictive accuracy, relevance, or overall model fit and data understanding.

For the Jester dataset, a distinct pattern emerged where metrics such as Tarwid, Pearson-II, Stiles, and Sokal-Sneath-V displayed the lowest MAE and highest R^2 values,

TABLE 13. MAE, P@k, and R^2 for MovieLens-100K.

Similarity metric	MAE	P@k	R^2
Sokal-Michener	0.830087	0.001696	0.348639
Sokal-Sneath-II	0.875721	0.001855	0.350687
Rogers-Tanimoto	0.782481	0.001882	0.436453
Faith	0.789872	0.053472	0.153221
Gower-Legendre	0.875721	0.001855	0.350687
Inner Product	0.830087	0.001696	0.348639
Russell-Rao	1.271247	**0.295413**	-1.108979
Forbes-I	1.300062	0.220042	-1.247775
Tarwid	0.977414	0.220042	-0.349657
Dennis	0.769628	0.057635	0.108965
Sokal-Sneath-IV	**0.221852**	0.044379	-0.063575
Gower	0.754356	0.152359	0.170132
Pearson-I	1.088808	0.222242	-0.674570
Pearson-II	0.922517	0.222242	-0.258467
Pearson-III	0.982833	0.220652	-0.392253
Pearson-Heron-I	0.762262	0.164395	0.067594
Pearson-Heron-II	1.032701	0.029639	-0.297233
Stiles	0.909078	0.001882	0.370750
Yule-Q-S	0.960129	0.221341	-0.350895
Yule-w	0.991869	0.221314	-0.450903
Disperson	0.769628	0.057635	0.108965
Hamann	0.818491	0.001696	0.363071
Michael	0.796702	0.208854	-0.000354
Baroni-Urbani-Buser-I	0.918947	0.001219	0.302441
Baroni-Urbani-Buser-II	0.917515	0.001219	0.304933
Peirce	1.045905	0.174973	-0.276769
Eyraud	1.561028	0.224231	-2.053843
Tarantula	0.293700	0.000159	0.077326
Ample	0.293700	0.000159	0.077326
Fossum	1.342698	0.221314	-1.397142
Forbes-II	0.919667	0.001882	0.392138
Vari	0.971982	0.001564	0.158180
Yule-Q-D	0.960129	0.221341	-0.350895
Shape-Difference	0.740371	0.001352	**0.558813**
Pattern-Difference	0.872803	0.218478	-0.105570
Mean-Manhattan	0.830087	0.001696	0.348639

TABLE 14. MAE, P@k, and R^2 for MovieLens-1M.

Similarity metric	MAE	P@k	R^2
Sokal-Michener	1.123213	0.000529	-0.880014
Sokal-Sneath-II	0.929902	0.000716	-0.275617
Rogers-Tanimoto	1.340046	0.000475	-1.317264
Faith	0.819032	0.025980	0.151463
Gower-legendre	0.929902	0.000716	-0.275617
Inner Product	1.112066	0.000538	-0.836397
Russell-Rao	1.218886	**0.249822**	-1.448582
Forbes-I	1.176707	0.214908	-1.243345
Tarwid	1.104606	0.214908	-1.004510
Dennis	8.632414	0.027955	-5682.515459
Sokal-Sneath-IV	**0.229868**	0.000438	0.015780
Gower	0.793196	0.059884	**0.187176**
Pearson-I	1.996752	0.200095	-4.651606
Pearson-II	0.970565	0.200095	-0.594337
Pearson-III	27.507256	0.019259	-466196.0767
Pearson-Heron-I	8.309241	0.047644	-5823.304107
Pearson-Heron-II	1.006575	0.014805	-0.216695
Stiles	1.008725	0.200095	-0.695151
Yule-Q-S	10.377394	0.104387	-8982.475772
Yule-w	20.964121	0.105376	-1591691.197
Disperson	8.632452	0.027955	-5682.606313
Hamann	1.591442	0.000533	-3.072908
Michael	9.832618	0.044875	-23709.00491
Baroni-Urbani-Buser-I	0.820826	0.000260	0.163236
Baroni-Urbani-Buser-II	0.850716	0.000260	0.163256
Peirce	1.885872	0.212504	-4.139985
Eyraud	2.111915	0.214656	-5.432240
Tarantula	0.244833	2.06954E-05	-0.013954
Ample	0.244833	2.06954E-05	-0.013954
Fossum	1.200955	0.215695	-1.321757
Forbes-II	0.916144	0.002566	-0.076831
Vari	1.147812	0.000467	-0.911535
Yule-Q-D	10.377394	0.104387	-8982.475772
Shape-Difference	0.916241	0.000173	0.099647
Pattern-Difference	0.925981	0.214838	-0.486057
Mean-Manhattan	1.123213	0.000529	-0.880014

TABLE 15. MAE, P@k, and R^2 for MovieLens-10M.

Similarity metric	MAE	P@k	R^2
Sokal-Michener	0.715741	0.000615	0.075463
Sokal-Sneath-II	0.700880	0.000714	0.152700
Rogers-Tanimoto	0.801168	0.000526	-0.026471
Faith	0.696601	0.044235	0.130541
Gower-Legendre	0.700880	0.000714	0.152700
Inner Product	0.700014	0.000615	0.096355
Russell-Rao	1.243323	**0.248301**	-1.430222
Forbes-I	1.706516	0.185442	-3.218441
Tarwid	1.005042	0.185442	-0.623106
Dennis	1.654526	0.034602	-298.832851
Gower	0.699287	0.101035	0.156115
Pearson-I	0.781337	0.000136	**0.391587**
Stiles	0.776793	0.000140	0.391314
Yule-w	1.345838	0.169983	-1267.226189
Disperson	1.654529	0.034602	-298.836804
Hamann	0.768185	0.000618	-0.124218
Baroni-Urbani-Buser-II	0.708379	0.000280	0.192061
Peirce	1.544580	0.152802	-2.372139
Eyraud	2.158879	0.188046	-5.557427
Tarantula	**0.111757**	2.73373E-05	0.021107
Ample	**0.111757**	2.73373E-05	0.021107
Fossum	1.328023	0.186652	-1.669630
Forbes-II	0.667105	0.000228	0.290489
Vari	0.884406	0.000512	-0.089978
Shape-Difference	0.517677	0.000194	0.250004
Pattern-Difference	0.835587	0.185917	-0.183838
Mean-Manhattan	0.715741	0.000615	0.075463

indicating good predictive accuracy. However, these metrics were less effective in terms of ranking relevance, as evidenced by their low P@k scores [45], [46]. In contrast, metrics like Dennis and Disperson achieved the highest P@k scores, highlighting their capability in recommending the most relevant items. Metrics such as Peirce, Yule-Q-S, and Yule-Q-D, however, performed poorly in terms of R^2, indicating challenges in model accuracy for this dataset.

The Epinions dataset presented a diverse range of performance across similarity metrics. Pearson-I, Peirce, Michael, and Russell-Rao emerged as leading metrics in terms of P@k, despite their less favorable MAE and R^2 scores. This suggests that these metrics are particularly useful for relevance-based recommendations, as in RSs, precision is often more critical than predictive errors.

The results of the analysis of the similarity metrics for the MovieTweetings and Electronics datasets showed that certain metrics exhibit similar behavior in both datasets. In particular, the metrics that perform best in the MovieTweetings dataset, such as Russell-Rao, Forbes-I, Peirce, also score well in the Electronics dataset. It can be inferred that these datasets likely exhibit similar patterns and trends. Moreover, the identification of similarity metrics that demonstrate strong performance across various datasets from different domains could yield insights concerning the choice of appropriate similarity metrics for building a cross-domain recommender

TABLE 16. MAE, P@k, and R^2 for Jester.

Similarity metric	MAE	P@k	R^2
Sokal-Michener	3.139624	0.013868	-0.255055
Sokal-Sneath-II	2.728853	0.072954	0.154898
Rogers-Tanimoto	2.695785	0.014918	0.068252
Faith	11.049162	0.013025	-13.252146
Gower-Legendre	2.728853	0.072954	0.154898
Inner Product	3.139624	0.013868	-0.255055
Russell-Rao	4.007103	0.076130	-0.151876
Forbes-I	3.790694	0.075970	-0.070742
Tarwid	**2.092937**	0.013037	0.251526
Dennis	3.168473	**0.076414**	-0.022993
Sokal-Sneath-IV	2.307972	0.013035	0.230664
Gower	3.576861	0.013041	-0.878985
Pearson-I	3.875026	0.013032	-0.576847
Pearson-II	2.104398	0.013030	**0.252725**
Pearson-III	2.872123	0.013033	0.013656
Pearson-Heron-I	3.526177	0.013029	-0.733516
Stiles	2.187976	0.013032	0.249323
Yule-Q-S	125.890682	0.016234	-599855.2728
Disperson	3.168621	**0.076414**	-0.022992
Hamann	3.135976	0.013868	-0.252865
Michael	2.354990	0.023370	0.068721
Peirce	66.080350	0.024916	-10572.48172
Eyraud	6.887838	0.013034	-3.696186
Tarantula	3.074490	0.076235	0.076732
Ample	3.085393	0.076235	0.084069
Fossum	3.878605	0.075993	-0.111984
Forbes-II	2.752836	0.069219	0.124317
Sokal-Sneath-V	2.298013	0.013033	-0.011510
Vari	3.066973	0.075281	0.127721
Yule-Q-D	125.788217	0.016234	-596819.205
Shape-Difference	13.534079	0.075973	-11.184028
Pattern-Difference	7.140724	0.013047	-3.907594
Mean-Manhattan	3.139624	0.013868	-0.255055

TABLE 17. MAE, P@k, and R^2 for Epinions.

Similarity metric	MAE	P@k	R^2
Sokal-Michener	8.552530	0.001138	-1257.817052
Sokal-Sneath-II	2.027112	0.000702	-3.661876
Rogers-Tanimoto	2.341814	0.000720	-11.127522
Faith	21.149303	0.001005	-15806.65648
Gower-Legendre	2.027112	0.000701	-3.661876
Inner Product	8.728164	0.001138	-1258.546065
Russell-Rao	3.657327	0.004817	-7.926313
Forbes-I	3.170117	0.003281	-8.402953
Tarwid	1.392492	0.001119	-3.612617
Dennis	3670.331719	0.001839	-64248062.44
Sokal-Sneath-IV	0.465062	0.000132	-0.490245
Gower	0.439463	0.000189	-1.035954
Pearson-I	1.937170	**0.009066**	-1.790181
Pearson-II	0.468127	0.002541	-0.407449
Pearson-III	0.358602	0.001327	-0.505461
Pearson-Heron-I	0.459113	0.001631	-0.539206
Pearson-Heron-II	0.428364	0.000417	-1.356647
Stiles	0.468124	0.002541	-0.407483
Yule-Q-S	4.242777	0.003338	-19.045252
Yule-w	0.469152	0.002541	**-0.405259**
Disperson	3909.35152	0.001764	-67861931.29
Hamann	4.395274	0.001270	-164.894685
Michael	34915.17085	0.005595	-14008368798
Baroni-Urbani-Buser-I	0.434108	9.4847E-05	-1.229076
Baroni-Urbani-Buser-II	0.396491	0.000113	-1.151803
Peirce	1.870563	0.006316	-1.519604
Eyraud	39.307852	0.000872	-7082.237446
Tarantula	**2.2245E-16**	1.89681E-05	0.000000
Ample	**2.2245E-16**	1.89681E-05	0.000000
Fossum	3.238732	0.001157	-11.883301
Forbes-II	1.817834	0.002807	-3.259539
Sokal-Sneath-V	0.600277	7.58725E-05	-3.943655
Vari	1.726803	0.000682	-7.914517
Yule-Q-D	4.242777	0.003338	-19.045252
Shape-Difference	0.651108	0.000625	-1.856209
Pattern-Difference	1.525874	0.001157	-2.914091
Mean-Manhattan	8.552530	0.001138	-1257.817052

system. Such systems leverage user interactions across different domains (i.e., books, movies, etc.). In the case of collaborative filtering-based cross-domain recommender systems, the choice of similarity metrics that are efficient across different domains is crucial. Moreover, comparing similarity metrics across different datasets can aid in identifying robust and stable metrics that consistently produce reliable results. This comparative analysis helps in selecting appropriate combinations of similarity metrics to be used in building ensemble-based recommender systems.

Russell-Rao consistently excelled in P@k, demonstrating a reliable capacity to identify relevant items across various datasets. However, it did not always provide the best predictive accuracy or model fit. A similarity metric may perform well in one aspect but may not necessarily deliver optimal performance in others, highlighting the inherent trade-offs in similarity metric selection. The selection must be context-dependent and aligned with the specific goals of the recommender system, where balancing objectives becomes crucial. For example, increasing precision might come at the expense of other metrics. Practitioners need to carefully weigh these trade-offs to ensure that the chosen similarity metric supports the targeted outcome. Thus, selecting a similarity metric is a multi-dimensional decision similar to those faced in multi-objective recommender systems [47].

The analysis also revealed two distinct groups of similarity metrics for each dataset: those that consistently produce the highest scores and those that perform poorly. This classification, detailed in Table 12, emphasizes that the success of a RS is strongly influenced by the selected similarity measures.

In future work, we will simulate the similarity metrics without negative co-occurrence (the parameter d, as mentioned in Table 2) and compare the results.

VI. CONCLUSION AND FUTURE WORK

In summary, this study delves into the impact and influence of various similarity metrics utilizing negative co-occurrence on the performance of RSs. We provide a comprehensive overview of these metrics in the context of implicit ratings and evaluate their effectiveness through experimental studies using seven well-known datasets. Our findings indicate that the performance of similarity metrics varies depending on the dataset's characteristics, such as sparsity or density, and the evaluation metrics used. Specifically, the Russell-Rao similarity metric consistently excelled in P@k across most datasets, suggesting its strength in relevance-based tasks despite not always achieving high predictive accuracy.

TABLE 18. MAE, P@k, and R^2 for MovieTweetings.

Similarity metric	MAE	P@k	R^2
Sokal-Michener	2.914475	0.001385	-28.952310
Sokal-Sneath-II	2.756885	0.000980	-2.201224
Rogers-Tanimoto	1.428518	0.000860	-0.399130
Faith	2.707855	0.002062	-8.351591
Gower-Legendre	2.756885	0.000980	-2.201224
Inner Product	2.912483	0.001381	-22.132197
Russell-Rao	6.860635	**0.027965**	-16.950614
Forbes-I	6.495003	0.016669	-19.967359
Tarwid	996.484062	0.016669	-559169.1928
Dennis	287.169795	0.002144	-391628.3996
Sokal-Sneath-IV	0.466479	0.000401	-0.328202
Gower	0.363666	0.000989	-0.042692
Pearson-I	1.433228	0.019296	-0.451485
Pearson-II	0.384355	0.005241	-0.149596
Pearson-III	0.387381	0.005071	-0.192950
Pearson-Heron-I	0.400812	0.002597	-0.212517
Pearson-Heron-II	0.425464	0.001345	-0.270182
Stiles	0.383967	0.005241	-0.149136
Yule-Q-S	31.740474	0.009270	-34891.57571
Yule-w	0.425799	0.004931	-0.652081
Disperson	277.107624	0.002129	-388383.6646
Hamann	1.929580	0.001706	-1.181332
Michael	99114.95728	0.007896	-8.11912E+11
Baroni-Urbani-Buser-I	**0.189984**	0.000247	**0.133843**
Baroni-Urbani-Buser-II	0.210572	0.000216	0.133747
Peirce	1.483627	0.016977	-0.529951
Eyraud	3297.721678	0.011283	-6120736.592
Fossum	5.687782	0.008322	0.000000
Forbes-II	1.526537	0.012655	-0.602405
Sokal-Sneath-V	0.323159	9.27644E-05	0.0299503
Vari	1.541977	0.001276	-0.280513
Yule-Q-D	31.740474	0.009270	-34891.57571
Pattern-Difference	1.471419	0.013312	-0.250029
Mean-Manhattan	2.914475	0.001385	-28.952310

TABLE 19. MAE, P@k, and R^2 for Electronics.

Similarity metric	MAE	P@k	R^2
Sokal-Michener	0.748712	0.000196	-0.135395
Sokal-Sneath-II	0.831576	0.000178	-0.169444
Rogers-Tanimoto	0.721913	0.000206	-0.134839
Faith	0.774720	0.000260	-0.120854
Gower-Legendre	0.831576	0.000178	-0.169444
Inner Product	0.747356	0.000194	-0.163672
Russell-Rao	4.105240	**0.011112**	-15.345139
Forbes-I	2.418617	0.006021	-6.018218
Tarwid	76.683504	0.001549	0.000000
Dennis	2.318588	0.001221	-15.806573
Sokal-Sneath-IV	**0.447297**	0.000645	-0.938746
Gower	0.769377	0.000350	-5.051139
Pearson-I	0.854021	0.003982	-0.810075
Pearson-II	0.842326	0.003982	-0.809696
Pearson-III	1.027304	0.002085	-4.177698
Pearson-Heron-I	1.567891	0.003200	-9.016673
Pearson-Heron-II	0.887058	0.001670	-0.406985
Stiles	0.837688	0.003982	-0.810227
Yule-Q-S	1.357159	0.004848	-10.261009
Yule-w	1.323491	0.004834	-6.061174
Disperson	2.280956	0.001234	-14.328572
Hamann	0.748527	0.000212	-4.667871
Michael	1.353982	0.004811	-6.640565
Baroni-Urbani-Buser-I	0.966231	0.000254	-4.823028
Baroni-Urbani-Buser-II	1.020022	0.000221	-5.582954
Peirce	0.837258	0.004965	-0.976537
Eyraud	32.126883	0.001545	0.000000
Forbes-II	90.889670	0.000141	0.000000
Sokal-Sneath-V	0.670059	0.000159	**-0.004092**
Vari	0.842054	0.000375	-5.034169
Yule-Q-D	1.357159	0.004848	-10.261009
Shape-Difference	1.838191	8.4877E-05	0.000000
Pattern-Difference	0.786683	0.005949	-0.827624
Mean-Manhattan	0.748712	0.000196	-0.135395

Additionally, our comparative analysis aims to guide users and recommender system developers in selecting the most effective similarity metrics for their specific use cases. One key contribution is the ability to select the most appropriate similarity metric for a given domain. These insights can help develop more adaptive and personalized recommender systems across various domains. Moreover, it highlights the importance of exploring a comprehensive range of similarity metrics beyond traditional ones.

The study also highlights that various similarity metrics demonstrated low MAE in certain datasets which may not always align with high P@k and R^2 scores, pointing to a trade-off between predictive accuracy and recommendation relevance.

The performance of a similarity metric is influenced by the characteristics of the data, including its sparsity or density, as well as the specific performance metrics used. Generally, no single similarity metric is universally optimal across all datasets; different metrics perform better on different datasets, aligning with the "no free lunch" theory [48]. Additionally, the analysis of datasets such as MovieLens, Jester, Epinions, MovieTweetings, and Electronics indicates that certain metrics can perform well across multiple domains. This suggests the potential for developing cross-domain recommender systems and underscores the need for a context-specific selection of similarity metrics. These findings highlight the value of using ensemble-based recommender systems that can combine the strengths of various metrics to enhance overall performance.

Future research should explore additional similarity metrics, particularly those without negative co-occurrence, and test them with a broader range of datasets to validate these findings.

Although a considerable number of similarity metrics have been tested in this study, there are still additional metrics that need to be explored and evaluated, particularly similarity metrics without negative co-occurrence. It is important to further test and evaluate these metrics to determine which metrics are the most effective for RSs.

Moreover, the datasets used in this work may not be sufficient to draw meaningful conclusions about the effectiveness of different similarity metrics; therefore, a larger number of datasets with a variety of data types is needed to make the analysis stronger.

Finally, the investigation in this study was limited to the use of the SAR algorithm. It is possible that the results and conclusions drawn from the performance of the similarity metrics will vary if a different algorithm is used. Since the complexity of SAR is $O(n^3)$, it would be more advisable to test other algorithms with lower complexity.

APPENDIX

See Tables 13–19.

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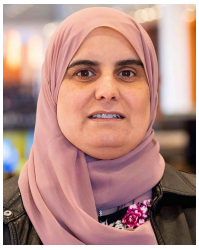
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